

2-D MOSFET Modeling Including Surface Effects and Impact Ionization by Self-Consistent Solution of the Boltzmann, Poisson, and Hole-Continuity Equations

Wenchao Liang, Neil Goldsman, Isaak Mayergoyz, *Fellow, IEEE*, and Phil J. Oldiges

Abstract—We present a new two-dimensional (2-D) MOSFET simulation method achieved by directly solving the Boltzmann Transport equation for electrons, the Hole-Current Continuity equation, and the Poisson equation self-consistently. The spherical harmonic method is used for the solution of the Boltzmann equation. The solution directly gives the electron distribution function, electrostatic potential, and the hole concentration for the entire 2-D MOSFET. Average quantities such as electron concentration and electron temperature are obtained directly from the integration of the distribution function. The collision integral is formulated to arbitrarily high spherical harmonic order, and new collision terms are included that incorporate effects of surface scattering and electron-hole pair recombination/generation. I-V characteristics, which agree with experiment, are calculated directly from the distribution function for an LDD submicron MOSFET. Electron-hole pair generation due to impact ionization is also included by direct application of the collision integral. The calculations are efficient enough for day-to-day engineering design on workstation-type computers.

I. INTRODUCTION

IN THIS paper, we present a new efficient two-dimensional (2-D) MOSFET simulation tool which is based on the self-consistent solution of the Boltzmann, Poisson, and Hole-Continuity equations.

To facilitate the design of deep-submicron devices, new modeling approaches, which are based on the solution of the Boltzmann Transport equation (BTE) [1]–[5], are being sought to complement existing techniques. The spherical harmonic (SH) method for solving the BTE represents a new “middle of the road” device design option. It gives more information than the Hydrodynamic model [6]–[11], while requiring only slightly more computation time. It provides information which is similar to spherical-band Monte Carlo calculations, and is computationally many times faster. The SH method can provide the momentum distribution function, using spherical band models [12], [13], as well as average quantities, for an

entire device. It is therefore a promising approach for modeling hot-electron phenomena, including MOSFET reliability, EPROM programming and impact-ionization, which depend on the shape of the distribution function. It has been shown to agree with Monte Carlo simulations which employ the same input parameters [14]–[16], while requiring orders of magnitude less CPU time to evaluate, and without statistical noise. The SH method has been employed to solve the BTE in one-dimensional (1-D) for BJT's as a post-processor [17], and self-consistently with the Poisson equation for simple 1-D test structures [18]. It has also been used as a post-processor in a first step to achieving 2-D MOSFET simulation [19], [20]. In addition, a generalized spherical harmonic approach has been achieved for the left-hand side (LHS) of the Boltzmann equation, in 1-D and 2-D, which allows the incorporation of an arbitrary number of spherical harmonics [20], [21].

Here, we further advance the SH approach, and demonstrate the viability of the method, by developing a stand-alone simulator for deterministic, self-consistent MOSFET modeling. We begin by introducing the device model which consists of the BTE for electrons, the Poisson equation, and Hole-Current Continuity equation. We then transform the LHS of the BTE into a tractable form using the generalized SH approach given in [20]. Next, we extend this approach to also express the collision integral to arbitrarily high spherical harmonic order in a general way. We also introduce new collision terms into the model to account for surface scattering in MOSFET inversion layers, Shockley–Read–Hall (SHR) electron-hole pair recombination-generation, and electron-hole pair generation resulting from impact ionization. To facilitate obtaining a numerical solution, we developed a new Scharfetter–Gummel-type discretization for the BTE. We also develop an iterative method to self-consistently solve the entire nonlinear device model, while simultaneously overcoming computational restrictions imposed by the three-dimensional (3-D) BTE. The simulation results directly provide the electron distribution function, electric potential and hole concentration for the entire MOSFET. To our knowledge, this is the first 2-D MOSFET self-consistent simulation using the SH approach that has been reported.

To help demonstrate the accuracy of the distribution function obtained by solving the BTE, we use it to simulate

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W. Liang, N. Goldsman, and I. Mayergoyz are with the Department of Electrical Engineering, University of Maryland, College Park, MD 20742 USA.

P. J. Oldiges is with Digital Equipment Corporation, Hudson, MA 01749 USA.

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current–voltage characteristics for an actual submicron device. For the first time, the SH method is shown to provide excellent agreement with experimental current–voltage values in both the subthreshold and above-threshold region. In addition to current density, from the distribution function we can readily extract other relevant quantities, including carrier concentration, electron average velocity, temperature and electron-hole pair generation for the entire MOSFET. To calculate the MOSFET distribution function for energies up to 3 eV requires less than one CPU min on a DEC alpha station when used as a post processor, and approximately 1 h for a full self-consistent 10-point I–V curve. The computation time is thus similar to the hydrodynamic model, and orders of magnitude faster than similar Monte Carlo simulations.

II. THE DEVICE MODEL

We begin our discussion by stating the device model that consists of the Poisson equation (1), the Boltzmann Transport equation for electrons (2), and the Current Continuity equation for holes (3)

$$\nabla_r^2 \phi(\vec{r}) = \frac{q}{\epsilon_s} \left[\frac{1}{4\pi^3} \int f(\vec{r}, \vec{k}) d\vec{k} - p(\vec{r}) + N_A(\vec{r}) - N_D(\vec{r}) \right] \quad (1)$$

$$\frac{1}{\hbar} \nabla_k \epsilon \cdot \nabla_r f(\vec{r}, \vec{k}) + \frac{q}{\hbar} \nabla_r \phi(\vec{r}) \cdot \nabla_k f(\vec{r}, \vec{k}) = \left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{\text{coll}} \quad (2)$$

$$\nabla_r \cdot [\mu_p p(\vec{r}) \nabla_r \phi(\vec{r}) + \mu_p V_t \nabla_r p(\vec{r})] = R_p - G_{p,\text{ii}} \quad (3)$$

where

$\phi(\vec{r})$	potential;
$f(\vec{r}, \vec{k})$	electron distribution function;
$\vec{k}$	electron wave-vector;
$\vec{r}$	position vector;
$p(\vec{r})$	hole concentration;
$N_D(\vec{r})$	doping concentration for donors
$N_A(\vec{r})$	doping concentration for acceptors;
R_p	net Shockley–Hall–Read (SHR) hole recombination rate;
$G_{p,\text{ii}}$	hole generation rate due to impact ionization;
$V_t = K_B T/q$; ϵ_s	dielectric constant of silicon;
μ_p	hole mobility;
$\hbar$	Planck's constant.

The collision term on the right-hand side (RHS) of the BTE represents acoustic, optical, and intervalley phonon scattering, impact ionization, ionized impurity scattering, MOSFET surface scattering, and electron-hole recombination.

A. Formulation of the Boltzmann Transport Equation

To make the BTE tractable, we begin with the approach put forth in [20], where the BTE was expressed in terms of an SH expansion to arbitrarily high order. However, this generalized method was limited to the special cases of optical, intervalley and acoustic phonon scattering. Here, we extend the arbitrarily high order formulation to also include a more

generalized collision integral. With a generalized collision-integral formulated, we use it to account for the effects of acoustic, intervalley, and optical phonon scattering, ionized impurity scattering, impact ionization, MOSFET surface roughness scattering, surface acoustic phonon scattering, and electron-hole recombination-generation.

1) *Left-Hand Side of BTE*: We begin the formulation of the BTE by expressing the distribution function in terms of the following SH expansion:

$$f(\vec{r}, \vec{k}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l f_l^m(\vec{r}, \epsilon) Y_l^m(\theta, \psi). \quad (4)$$

Here, $Y_l^m(\theta, \psi)$ are the well-known spherical harmonic basis functions [22], θ and ψ are the polar and azimuthal angles of the wave-vector, respectively. $f_l^m(\vec{r}, \epsilon)$ represents the expansion coefficients. (Throughout this paper, unless explicitly stated, the summation will be assumed to be from $l = 0$ to $l = \infty$, with $-l < m < l$.) The eventual goal of the approach is to determine the coefficients and thereby the distribution function for the MOSFET. It was shown in [20], that by substituting the spherical harmonics expansion for the distribution function into the BTE, while taking advantage of the recurrence and orthogonality relationships between spherical harmonics, we can generalize the LHS of the BTE, and obtain the following system of equations for the unknown coefficients [20]:

$$\begin{aligned} & \sum_{i=1}^2 \left\{ v(\epsilon) \left[\frac{\partial}{\partial x_i} - q E_i(\vec{r}) \left(\frac{\partial}{\partial \epsilon} - \frac{l-1}{2} \frac{\gamma'(\epsilon)}{\gamma(\epsilon)} \right) \right] \hat{a}_i^+ \right. \\ & \quad \left. + v(\epsilon) \left[\frac{\partial}{\partial x_i} - q E_i(\vec{r}) \left(\frac{\partial}{\partial \epsilon} + \frac{l+2}{2} \frac{\gamma'(\epsilon)}{\gamma(\epsilon)} \right) \right] \hat{a}_i^- \right\} f_l^m \\ & = \left[\frac{\partial f}{\partial t} \right]_{\text{coll}}^m \end{aligned} \quad (5)$$

where $v(\epsilon) = \sqrt{2m\gamma(\epsilon)}/m\gamma'(\epsilon)$, $\gamma'(\epsilon) = d\gamma(\epsilon)/d\epsilon$, $\gamma(\epsilon)$ represents the dispersion relation, and $E_i(\vec{r})$ is the electric field in the i direction. The sum over i represents the Cartesian directions of a 2-D device cross section in real-space. The raising and lowering operators $\hat{a}_i^{\pm}$ which we developed to relate the coefficients are

$$\begin{aligned} \hat{a}_1^- f_l^m & \equiv \frac{1}{2} \{ -\alpha_{l-1}^m \alpha_l^m f_{l-1}^{m-1} + \alpha_{l-1}^{-m} \alpha_l^{-m} f_{l-1}^{m+1} \} \\ \hat{a}_1^+ f_l^m & \equiv \frac{1}{2} \{ \alpha_{l+1}^{-m+1} \alpha_l^{-m+1} f_{l+1}^{m-1} - \alpha_{l+1}^{m+1} \alpha_l^{m+1} f_{l+1}^{m+1} \} \\ \hat{a}_2^+ f_l^m & \equiv \alpha_{l-1}^{-m+1} \alpha_l^m f_{l-1}^m \\ \hat{a}_2^- f_l^m & \equiv \alpha_{l+1}^m \alpha_l^{-m+1} f_{l+1}^m; \quad \alpha_l^m = \sqrt{\frac{l+m}{2l+1}}. \end{aligned} \quad (5a)$$

2) *Generalized Collision Integral*: Before the above system can be solved, the collision integral must also be expressed in terms of spherical harmonics. As mentioned above, previously the collision integral was formulated to arbitrarily-high order for only special cases [20]. Here, we generalize the approach. We begin by substituting the SH expansion for

$f(\vec{r}, \vec{k})$ into the BTE's collision integral

$$\left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{\text{coll}} = \frac{V}{(2\pi)^3} \sum_{l,m,i} \int [f_l^m(\vec{r}, \epsilon') Y_l^m(\theta', \psi') S_i(\vec{k}', \vec{k}) - f_l^m(\vec{r}, \epsilon) Y_l^m(\theta, \psi) S_i(\vec{k}, \vec{k}')] d^3 \vec{k}' \quad (6)$$

where $S_i(\vec{k}', \vec{k})$ represents the i th type of transition probability per unit time. The summation over i represents the different scattering processes which occur in MOSFET's. Evaluation of the first term in (6) is especially difficult since it involves an integration over a product of two functions that both depend on the primed angle. To overcome this difficulty, we begin by expanding the transition probability in terms of a Legendre polynomial series

$$S_i(\vec{k}, \vec{k}') = \sum_{l=0}^{\infty} S_{l,i}(\epsilon, \epsilon') P_l(\cos \xi) \quad (7)$$

where ξ is the angle between $\vec{k}$ and $\vec{k}'$; ϵ , and ϵ' are the energies associated with $\vec{k}$ and $\vec{k}'$, respectively, and $S_{l,i}(\epsilon, \epsilon')$ is the l th Legendre coefficient of the transition probability, which is determined by projecting $S_i(\vec{k}, \vec{k}')$ onto the l th Legendre polynomial basis function. This form of the transition probability facilitates the use of addition theorem [22], which allows us to write (7) in terms of a spherical harmonic expansion as

$$S_i(\vec{k}, \vec{k}') = \sum_{l,m}^{\infty} \frac{4\pi}{2l+1} S_{l,i}(\epsilon, \epsilon') Y_l^m(\theta, \psi) Y_l^m(\theta', \psi'). \quad (8)$$

By substituting the above equation into the collision integral, integrating over θ', ψ' and using orthogonality, the collision integral for the i th scattering process immediately reduces to the following generalized expression:

$$\begin{aligned} \left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_i &= \frac{f_0^0(\vec{r}, \epsilon)}{\tau_{0,i}^{\text{in}\mp}(\epsilon)} Y_0^0(\theta, \psi) - \frac{f_0^0(\vec{r}, \epsilon)}{\tau_{0,i}^{\text{out}\mp}(\epsilon)} Y_0^0(\theta, \psi) \\ &- \sum_{l=1,m}^{\infty} \frac{f_l^m(\vec{r}, \epsilon)}{\tau_{0,i}^{\text{out}\mp}(\epsilon)} Y_l^m(\theta, \psi) \\ &+ \sum_{l=1,m}^{\infty} \left[\frac{1}{2l+1} \frac{V}{(2\pi)^9} \int f_l^m(\vec{r}, \epsilon') S_{l,i}(\epsilon', \epsilon) h(\epsilon') d\epsilon' \right] \\ &\cdot Y_l^m(\theta, \psi) \end{aligned} \quad (9)$$

where

$$\begin{aligned} \frac{1}{\tau_{0,i}^{\text{out}\mp}(\epsilon)} &= \frac{V}{(2\pi)^3} \int S_{0,i}^{\mp}(\epsilon, \epsilon') h(\epsilon') d\epsilon' \\ \frac{1}{\tau_{0,i}^{\text{in}\mp}(\epsilon)} &= \frac{V}{(2\pi)^3} \int S_{0,i}^{\mp}(\epsilon', \epsilon) h(\epsilon') d\epsilon'. \end{aligned} \quad (10)$$

In (9) and (10) we have introduced the density of states, $h(\epsilon) = \frac{4\sqrt{2\pi}}{\hbar^3} m^{*3/2} \gamma^{1/2}(\epsilon) \gamma'(\epsilon)$. The superscript out refers to a process where the electron is scattered out of the state $\vec{k}$ to

another state $\vec{k}'$, while the superscript in represents scattering into $\vec{k}$. The superscripts $\mp$ refer to absorption ($-$) and emission ($+$) of energy respectively. (For elastic processes $\mp$ have been eliminated, since there is no energy emission or absorption. In cases where there is no process for energy emission (absorption), the superscript $+$ ($-$) has been eliminated.)

Equation (9) reflects several important results. First of all, the scattering-out part of the collision integral, which are the negative terms in (9), as well as the zero order scattering-in term, have been transformed into algebraic expressions. This greatly simplifies the BTE. In addition, the coupling between different spherical harmonics has been eliminated so the formulated BTE can be solved independently of its angular distribution to arbitrary order. (This can be seen from the fact that the SH basis functions have been factored out of the integral, and there are no products between SH basis functions.) In essence, this transforms the BTE from a six-dimensional (6-D) equation into one that is solvable in four-dimensional (4-D), to arbitrary SH order. Finally, within the context of the spherical band approximation, for most of the collisions which occur in MOSFET's, (9) reduces to a simple relaxation-time form.

The specific formulation for each of the scattering terms is given in the Appendix. Surface roughness and 2-D acoustic scattering, as well as SHR recombination, have not been previously formulated in terms of the SH expansion, so we give a fairly detailed description of their derivations. Intervalley, optical and acoustic phonons, impact ionization, and ionized impurity scattering have been expressed in terms of the SH expansion previously [17], [20], [23], so in the Appendix we only briefly describe how they fit into our generalized collision integral.

3) *Final Form of BTE*: With the mathematical formalism of the scattering mechanisms achieved, we now must truncate the expansion in order to obtain a numerical solution. It has been demonstrated for both single and multiband dispersion relations that a low order truncation leads to results which agree with Monte Carlo calculations for the homogeneous and 1-D cases [14]–[16]. It can be shown that, due to the group properties of spherical harmonics [22], the truncation error, with respect to l , is the same for 2-D as in 1-D [20]. We therefore use an expansion which includes the first four spherical harmonics ($Y_0^0, Y_1^{-1}, Y_1^0, Y_1^1$). We next follow a procedure similar to that of [20], but also include the additional scattering mechanisms which are necessary for MOSFET simulation. With this procedure, first the state-of-the-art H transformation is employed [17], where the total energy H is used as the independent energy variable as opposed to the kinetic energy ϵ , ($H = \epsilon - q\phi$). This transformation has the effect of removing derivatives with respect to energy. (This simplification, however, comes at the price of a curvilinear simulation domain, which complicates the solution of the coupled Poisson-Boltzmann-Continuity system.) With the H transformation, the expansion coefficients are redefined to be $f_l^m(t, \vec{x}, \epsilon) \equiv F_l^m(t, \vec{x}, H)$. After applying symmetry arguments, and a considerable amount of algebra, we can arrive at the following tractable expression for the symmetrical part of

the distribution function throughout the device:

$$\begin{aligned}
& -\nabla \cdot \left[\frac{\gamma^{3/2}(H+q\phi)\tau(H+q\phi)}{\gamma'(H+q\phi)} \nabla F_0^0(x,y,H) \right] \\
& = \frac{3(\varrho-1)D_n^2 m^{*5/2}}{2\sqrt{2}\pi\hbar^3 \rho\omega_n} \left(\frac{\gamma^{1/2}(H+q\phi)\gamma'(H+q\phi)}{e^{\hbar\omega_n/K_0T} - 1} \right) \\
& \quad \cdot \left\{ \sqrt{\gamma(H+q\phi-\hbar\omega_n)}\gamma'(H+q\phi-\hbar\omega_n) \right. \\
& \quad \cdot [F_0^0(x,y,H-\hbar\omega_n) - e^{\hbar\omega_n/K_0T} F_0^0(x,y,H)] \\
& \quad + \sqrt{\gamma(H+q\phi+\hbar\omega_n)} \cdot \gamma'(H+q\phi+\hbar\omega_n) \\
& \quad \cdot [e^{\hbar\omega_n/K_0T} F_0^0(x,y,H+\hbar\omega_n) - F_0^0(x,y,H)] \left. \right\} \\
& \quad + \frac{3}{2} m^* \gamma^{1/2}(H+q\phi)\gamma'(H+q\phi) \\
& \quad \cdot \left[\frac{2F_0^0(x,y,2H+\Delta_{ii}+q\phi)}{\tau_{0,ii}^{\text{in}}(2H+2q\phi+\Delta_{ii})} - \frac{F_0^0(x,y,H)}{\tau_{0,ii}^{\text{out}}(H+q\phi)} \right. \\
& \quad + \frac{f_T}{\tau_{0,\text{rec}}^{\text{out}}(H+q\phi)} \exp\left(\frac{H-H_t}{K_B T_0}\right) \\
& \quad \left. - \frac{1-f_T}{\tau_{0,\text{rec}}^{\text{out}}(H+q\phi)} F_0^0(x,y,H) \right] \quad (11)
\end{aligned}$$

where

$$\begin{aligned}
\frac{1}{\tau} &= \frac{1}{\tau_{0,\text{op}}^{\text{out}}} + \frac{1}{\tau_{0,ii}^{\text{out}}} + \frac{1-f_T}{\tau_{0,\text{rec}}^{\text{out}}} + \frac{1}{\tau_{0,\text{ac}}^{\text{out}}} \\
&+ \frac{1}{\tau_{0,\text{sa}}^{\text{out}}} + \frac{1}{\tau_{0,\text{sr}}^{\text{out}}} + \frac{1}{\tau_{1,\text{imp}}^{\text{out}}}. \quad (12)
\end{aligned}$$

In (11), D_n = optical phonon deformation potential; ω_n = optical phonon vibrational frequency; F_0^0 = symmetrical part of electron distribution; Δ_{ii} = the threshold energy of impact ionization scattering; and ϱ = the number of equivalent valleys. f_T and H_t are the SHR trap occupation probability and energy, respectively. In (11) and (12) the subscripts op, ii, rec, ac, sa, sr, and imp represent optical, impact ionization, SHR recombination, acoustic, surface acoustic, surface roughness, and ionized impurity scattering, respectively. Explicit expressions for these τ 's and f_T are given in the Appendix.

Equation (11) is a self-adjoint difference-differential equation in three dimensions: (3-D), (2-D) real-space, and (1-D) energy-space, and thus, dimensionality has been reduced from the original five-dimensional (5-D) BTE of (2). Also, the collision integral has been analytically evaluated thereby significantly facilitating the BTE's solution. The LHS of (11) generally corresponds to the first two terms of the BTE shown in (2), while the RHS of the equation describes the effects of inelastic phonon scattering, impact ionization, and recombination-generation.

We solve (11) over the 3-D domain. The real-space domain is the 2-D MOSFET cross section, where we employ boundary conditions similar to those used in [20]. In energy-space we use the first two bands of $\gamma(\varepsilon)$ given by [12], [13]. This dispersion relation ranges from 0 eV to 3.02 eV. At $\varepsilon = 0$ and $\varepsilon_{\text{max}} = 3.02$ eV we take advantage of the fact that the density of states goes to zero. This leads to the boundary conditions which are reduced forms of (11).

B. Poisson and Hole-Continuity Equations

Within the context of our device model, the Poisson and Hole-Continuity equations are differential-integral equations, since they depend on ϕ, p and $\int f d\mathbf{k}$. To simplify these equations, we first integrate the distribution function to obtain the electron concentration $n(x, y)$. To achieve this, we directly map $F_0^0(x, y, H)$, which is obtained from the solution to the BTE, into energy-space to obtain $f_0^0(x, y, \varepsilon)$ and then integrate over the density of states to obtain $n(x, y)$. We then insert the values for $n(x, y)$ into the RHS of the Poisson equation, as well as the SHR recombination term of the Hole-Current-Continuity equation. The hole generation rate arising from impact ionization also depends on the distribution function. We extract the hole generation rate from the BTE's collision integral. By integrating the impact ionization term of the collision integral over momentum space we obtain the electron generation rate. Since the electron and hole generation rates are equal, this integration will also give the hole generation rate due to impact ionization $G_{p,ii}$. Mathematically, this can be expressed as

$$G_{p,ii} = \int \frac{F_0^0(x, y, H)}{\tau_{0,ii}^{\text{out}}(H+q\phi)} h(\varepsilon) d\varepsilon. \quad (13)$$

The explicit values for the impact ionization rate $1/\tau_{0,ii}$ are taken from [23].

III. METHOD OF NUMERICAL SOLUTION

We solve the Poisson equation, the BTE, and the Hole equation using a decoupled approach. The Poisson equation is directly discretized with finite differences and solved with sparse matrix Gaussian elimination. The Hole-Current-Continuity equation is first discretized using a Scharfetter-Gummel approach and then solved iteratively [24]. To help account for exponential variation in the distribution function without employing a large number of mesh-points, we also have developed a new Scharfetter-Gummel-type scheme for the BTE. To achieve this, we analytically solve the homogeneous part of (11) between mesh points, and account for the nonhomogeneous part on the mesh points. Employing finite differences then leads to the following discretized form of the BTE:

$$\begin{aligned}
& \left[\eta_{i+\frac{1}{2}jm} \frac{2(F_{i+1jm} - F_{ijm})}{h_i(h_i + h_{i-1})} - \eta_{i-\frac{1}{2}jm} \frac{2(F_{ijm} - F_{i-1jm})}{h_{i-1}(h_i + h_{i-1})} \right] \\
& + \left[\eta_{i,j+\frac{1}{2}m} \frac{2(F_{ij+1m} - F_{ijm})}{k_j(k_j + k_{j-1})} \right. \\
& \quad \left. - \eta_{i,j-\frac{1}{2}m} \frac{2(F_{ijm} - F_{ij-1m})}{k_{j-1}(k_j + k_{j-1})} \right] \\
& + \frac{3(\varrho-1)D_n^2 m^{*5/2}}{2\sqrt{2}\pi\hbar^3 \rho\omega_n} \left(\frac{\gamma_{ijm}^{1/2} \gamma'_{ijm}}{e^{\hbar\omega_n/K_0T} - 1} \right) \\
& \quad \cdot \left\{ \sqrt{\gamma_{ijm-l_n}} \gamma'_{ijm-l_n} \cdot [F_{ijm-l_n} - e^{\hbar\omega_n/K_0T} F_{ijm}] \right. \\
& \quad + \sqrt{\gamma_{ijm+l_n}} \gamma'_{ijm+l_n} [e^{\hbar\omega_n/K_0T} F_{ijm+l_n} - F_{ijm}] \left. \right\} \\
& - \frac{3}{2} m^* \gamma_{ijm}^{1/2} \gamma'_{ijm}
\end{aligned}$$

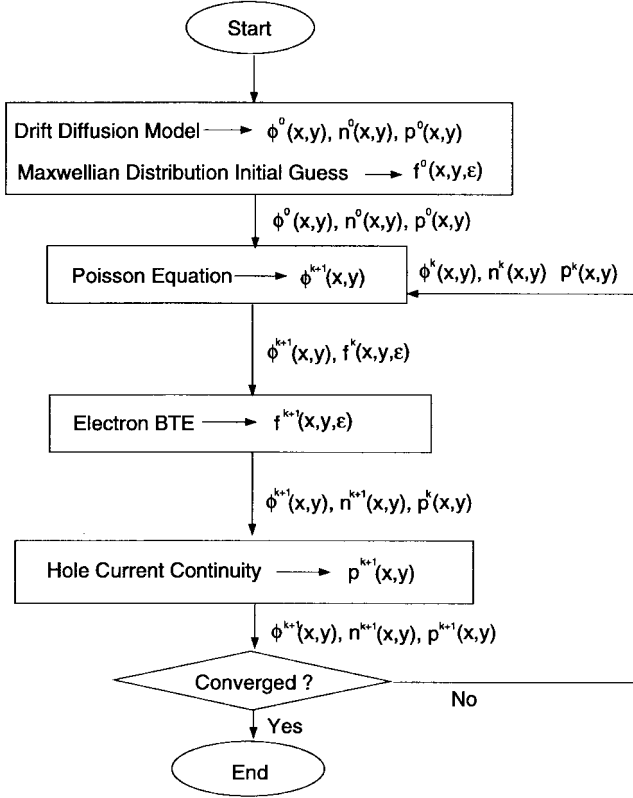


Fig. 1. The flow chart of the overall decoupled algorithm used to self-consistently solve the Poisson, Boltzmann, and Hole-Continuity equations for the 2-D MOSFET.

$$\begin{aligned}
 & \left[\frac{F_{ijm}}{\tau_{ii,ijm}} - \frac{2F_{ij2m+\Delta_{ii}}}{\tau_{ii,ij2m+\Delta_{ii}}} - \frac{f_T}{\tau_{rec,ijm}} \right. \\
 & \quad \cdot \exp\left(\frac{m\Delta H - H_i}{K_B T_0}\right) + \left. \frac{1 - f_T}{\tau_{rec,ijm}} F_{ijm} \right] \\
 & = 0
 \end{aligned} \tag{14}$$

where $F_{ijm} = F(x_i, y_j, m\Delta H)$; $l_n =$ closest integer to $\frac{n_{\omega n}}{\Delta H}$; $\gamma_{ijm} = \gamma(m\Delta H + q\phi(x_i, y_j))$; $\gamma'_{ijm} = \gamma(m\Delta H + q\phi(x_i, y_j))'$; the dispersion relation with respect to energy; $\eta_{i+\frac{1}{2}jm} = \frac{\zeta_{i+1jm} - \zeta_{ijm}}{\ln \zeta_{i+1jm} - \ln \zeta_{ijm}}$; and $\zeta = \frac{\gamma^{3/2} \tau}{\gamma'}$.

Since (14) is a finite difference equation in 3-D, it actually represents a system of $N_x \times N_y \times N_H \approx 100\,000$ discrete equations. Since this system is diagonally dominant, we efficiently solve it, and overcome constraints of 3-D simulation, with the conjugate gradient method. The whole coupled system of the Poisson, BTE, and Hole-Continuity equations is solved using a damped Gummel-type iterative approach. The flow chart in Fig. 1 shows the overall algorithm employed.

IV. RESULTS

We demonstrate the new simulator by modeling a deep sub-micron MOSFET where we obtain excellent agreement with experimental I-V values. The device is a 0.35- μm effective channel length LDD N-MOSFET [25]. It has maximum n^+ , n^- , and substrate doping on the order of 10^{20} , 10^{18} , and 10^{16} , respectively. The oxide thickness was 0.0096 μm , and source/drain junction depth of approximately 0.17 μm . Values for the transport parameters used in these calculations are

TABLE I

parameter	value	unit
D_{ac}	4.0	eV
D_n	5.0×10^8	eV/cm
E_{op}	0.05	eV
V_s	9.0×10^5	cm/sec
Γ	1.96	$\text{cm}^2/\text{V}^2\text{sec}$
n_{scr}	1.45×10^{15}	$1/\text{cm}^3$
D_{sa}	17.8	eV
δ	13.1	

given in Table I. It is worth noting that, where applicable, the transport parameter values we used are the same, or very similar to, the values employed in MOSFET Monte Carlo simulations [12]. For the case of surface scattering, it was not applicable to take values from [12] since we employed different transport models. Since these values are not well known, we chose those that gave the best fit to the experimental I-V characteristics. This lead to values for surface acoustic phonon deformation potential, and the product of rms step size and correlation length of: 17.8 eV and 5 $\text{\AA}^2$, respectively. We found that the values we chose are reasonable when compared to those put forth by others [26], [27]. We note that the transport parameters are not unique, and will be more closely determined only by continued experiment and calibration to specific fabrication processes.

To obtain an appreciation of the detailed physics of the device operation and carrier transport, we first show the simulation results for one specific bias point with $V_{ds} = 3$ V and $V_{gs} = 3$ V. We choose this bias because it represents the saturation region where 2-D effects are very important. The self-consistent solution gives electrostatic potential, hole concentration and the electron distribution function $f(\vec{r}, \vec{k})$ for the entire device. In Fig. 2 we show the resulting distribution function. However, since the distribution function is a 5-D quantity, plotting it is difficult. We instead show the resulting 3-D energy distribution functions along three different lines across the device. Fig. 2(a) shows the energy distribution function along the channel, 0.001 μm below the interface. The distribution function has the overall character one would expect, being Maxwellian in the source region, and very non-Maxwellian or heated in the drain side of the channel. Since the distribution function gives the number of electrons with a specific momentum everywhere in the device, it provides virtually a complete physical description of the device operation. For example, the distribution function near the drain will give the specific number of hot electrons with energy greater than 1.1 eV in that region. This information can then be used to quantify hot-electron reliability effects. Fig. 2(b) shows the energy distribution function along a line that is 0.2 μm below the interface. It is interesting to observe that LDD device structure is clearly reflected in the shape of the distribution function. Fig. 2(c) shows that the energy distribution function along a line deep in the substrate 0.64 μm below the interface. Since in this region, currents and fields are minimal, the resulting Maxwellian form is expected.

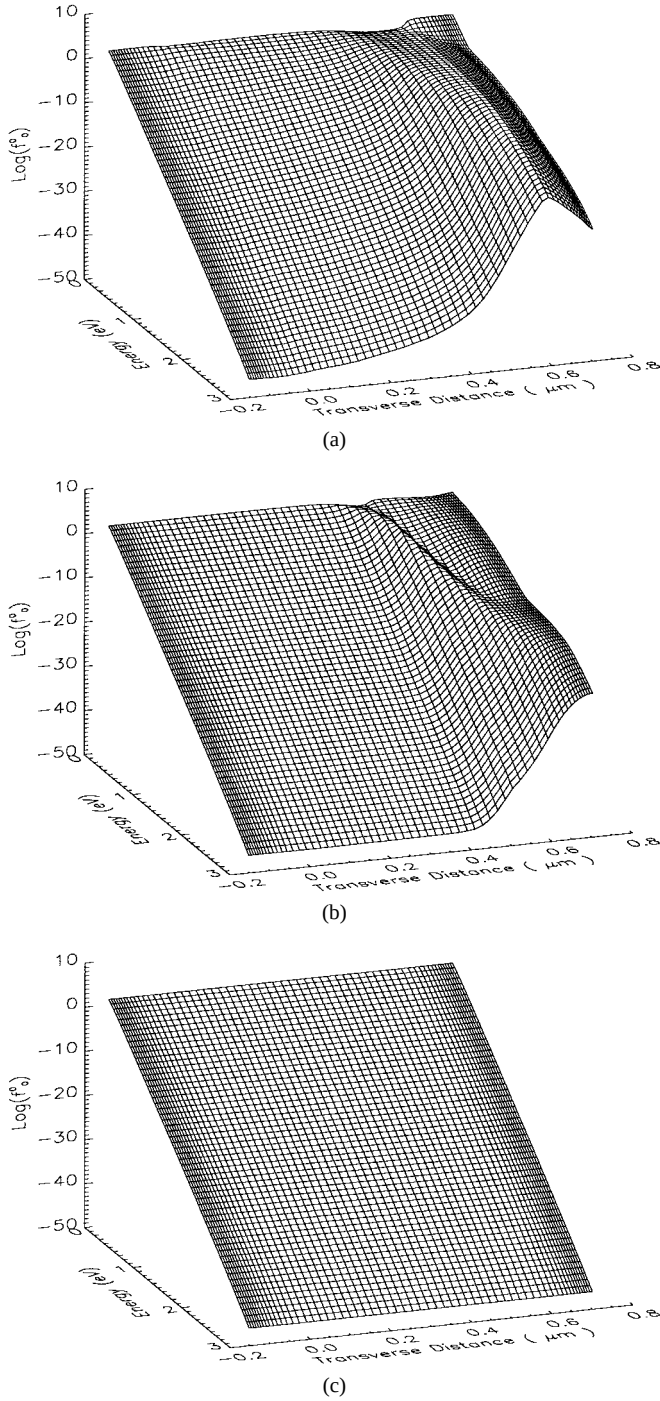


Fig. 2. (a) The distribution function along the channel $0.001 \mu\text{m}$ below the surface. (b) The distribution function along a line $0.2 \mu\text{m}$ below the surface. (c) The distribution function along a line $0.64 \mu\text{m}$ below the surface, which is in the substrate. The applied bias is $V_{GS} = 3 \text{ V}$ and $V_{DS} = 3 \text{ V}$.

In addition to the details of the distribution function, the SH method readily gives average values, or moments $\langle A(x, y) \rangle$, as well. These moments are calculated with the following expression, which can usually be integrated over θ and ϕ analytically:

$$\langle A(x, y) \rangle = \int A(k, \theta, \psi) \sum_{l=0}^{\infty} \sum_{m=-l}^l f_l^m(\vec{r}, \epsilon) \cdot Y_l^m(\theta, \psi) k^2 \sin \theta dk d\theta d\psi. \quad (15)$$

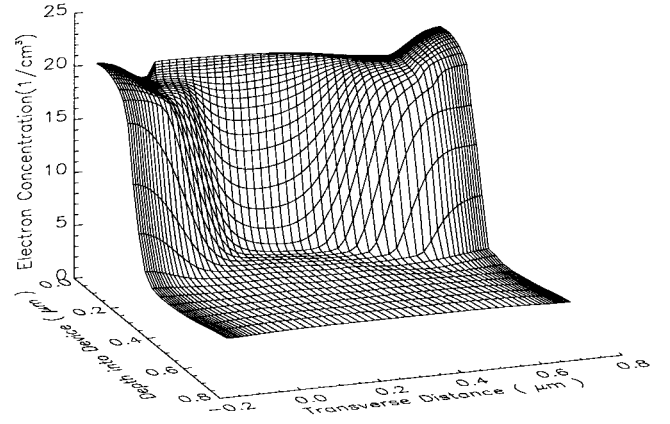


Fig. 3. The calculated electron concentration for an applied bias of $V_{GS} = 3 \text{ V}$ and $V_{DS} = 3 \text{ V}$. The electron concentration was obtained by integrating the calculated distribution function.

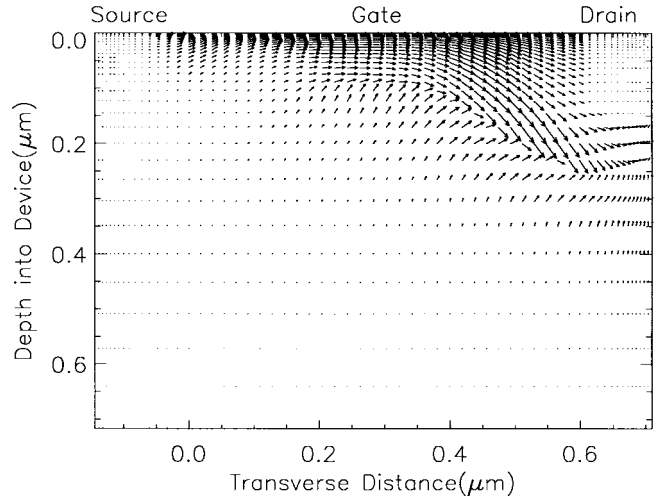


Fig. 4. The calculated electron average velocity for an applied bias of $V_{GS} = 3 \text{ V}$ and $V_{DS} = 3 \text{ V}$. The current density was obtained by taking the velocity moment of the distribution function.

In Fig. 3 we show the electron concentration, which is obtained by setting $A(k, \theta, \phi) = 1$, and performing the above integration. The figure clearly shows the onset of the pinch off effect. Fig. 4 shows the average electron velocity throughout the device, which is determined by integration while setting $A(k, \theta, \phi) = \vec{v}(k, \theta, \phi)$. The 2-D vector nature of electron flow is clearly demonstrated in this figure.

To help demonstrate the self-consistent nature of the solution to (1)–(3), we show the results for electrostatic potential and hole concentration in Figs. 5 and 6, respectively. In Fig. 6 it is interesting to observe the peak in the hole concentration near the drain, which is due to impact-ionization.

While the plots in Figs. 3–6 are in qualitative agreement with those of 2-D hydrodynamic and/or drift-diffusion simulators, Fig. 7 helps to demonstrate the utility of solving for the complete distribution function. In Fig. 7 we show contour plots of the electron temperature and electron-hole pair generation rate due to impact ionization on the same coordinate axes. We plot them this way to demonstrate that the peak temperature and peak generation rate do not coincide. It is worth noting

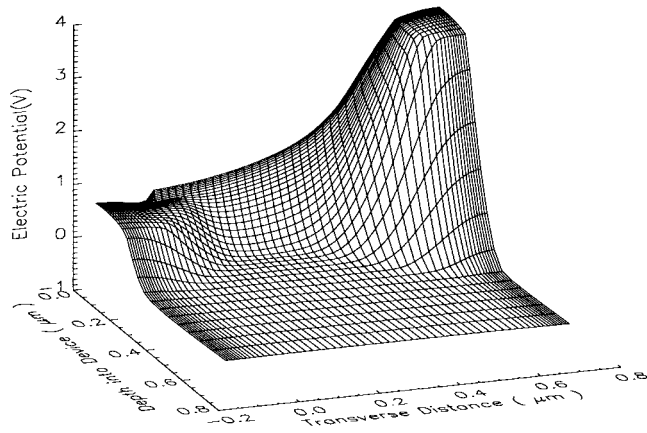


Fig. 5. The calculated electrostatic potential for an applied bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V. The potential was obtained by the self-consistent solution of the Poisson, Boltzmann, and Hole-Continuity equations.

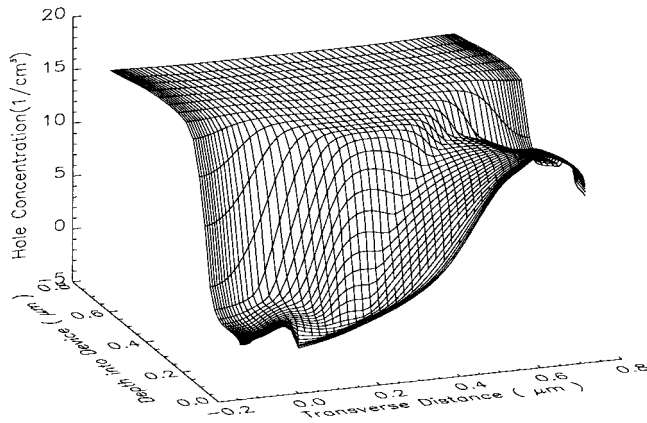


Fig. 6. The calculated hole concentration for an applied bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V. The hole concentration was obtained by the self-consistent solution of the Poisson, BTE, and Hole-Continuity equations.

that values for generation are not obtained empirically by mapping from space independent rates as is done in the drift-diffusion and hydrodynamic models, but directly from the Boltzmann equation by integrating the impact-ionization collision integral. The figure suggests that device design should be based on the distribution function, and not simply its averages.

As mentioned above, we used the SH method to calculate I-V characteristics, and demonstrate agreement with experiment for the first time. These I-V characteristics are not calculated with mobility models, as is done in the drift-diffusion and hydrodynamic approaches, but directly from basic transport phenomena such as scattering and band-structure. To calculate I-V characteristics, we take the velocity moment for a given bias, as shown in Fig. 4, and then integrate it over the surface at the source or drain contact. Fig. 8(a) shows the excellent agreement between experimental and simulated I-V characteristics calculated by solution of the BTE for the subthreshold region. The ability for a simulator to calculate subthreshold current is very important since this region of operation can provide information about the doping profile. In contrast with the Monte Carlo method, the SH approach has

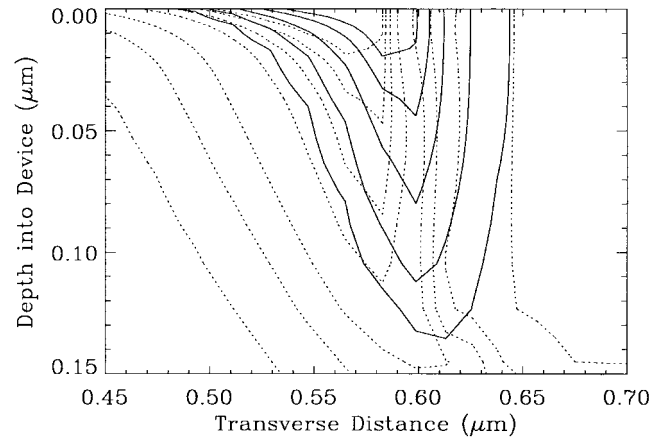


Fig. 7. Contour plots of the electron temperature (dashed lines) and the electron-hole pair generation rate due to impact ionization (solid lines). The contour values for electron temperature are 5000 K, 4800 K, 4500 K, 3000 K, 2000 K, 1000 K, and 500 K. The contour values for electron-hole pair generation rate are 2.3×10^{27} [1/cm³·s], 1.7×10^{27} [1/cm³·s], 1.0×10^{27} [1/cm³·s], 4.0×10^{26} [1/cm³·s], and 1.0×10^{26} [1/cm³·s].

no difficulty in solving the BTE in the subthreshold region of device operation. Fig. 8(b) shows the experimental and simulated I-V characteristics for the triode and saturation regions of device operation. Again, agreement between calculated and experimental values is excellent.

V. CONCLUSION

We have demonstrated a new 2-D MOSFET simulator. The simulator is based on the deterministic self-consistent solution of the BTE, Poisson, and Hole-Continuity equations. We applied the simulator to a 0.3-μm effective channel length 2-D MOSFET and obtained excellent agreement with experimental results for I-V characteristics. Since the approach requires only slightly more CPU time than the Hydrodynamic method, it is potentially useful for mainstream device simulation. It appears that its main advantage over the hydrodynamic (HD) model is that it provides the distribution function, from which most device characteristics can be derived, including I-V characteristics and hot-electron phenomena. Other apparent advantages are that it requires solving only three equations (as opposed to five for the HD), and it is based on specific scattering mechanism and band-structure (as opposed to HD mobility models extracted from empirical and Monte Carlo data). Since the simulator uses the spherical band approximation, which averages the effects of warped band-structures onto a sphere, it cannot provide as many details in 5-D phase-space as full-zone Monte Carlo calculations. However, it is many times faster than Monte Carlo simulations and free of statistical noise. It may thus be useful as a complement to detailed Monte Carlo simulations, or as a stand-alone simulator for day-to-day engineering problems where knowledge of the distribution function is required. The efficiency of the new simulator and agreement with experiment indicate the approach has considerable promise. Future work will better ascertain its predictive values and its part in the overall field of device simulation for MOSFET manufacturing.

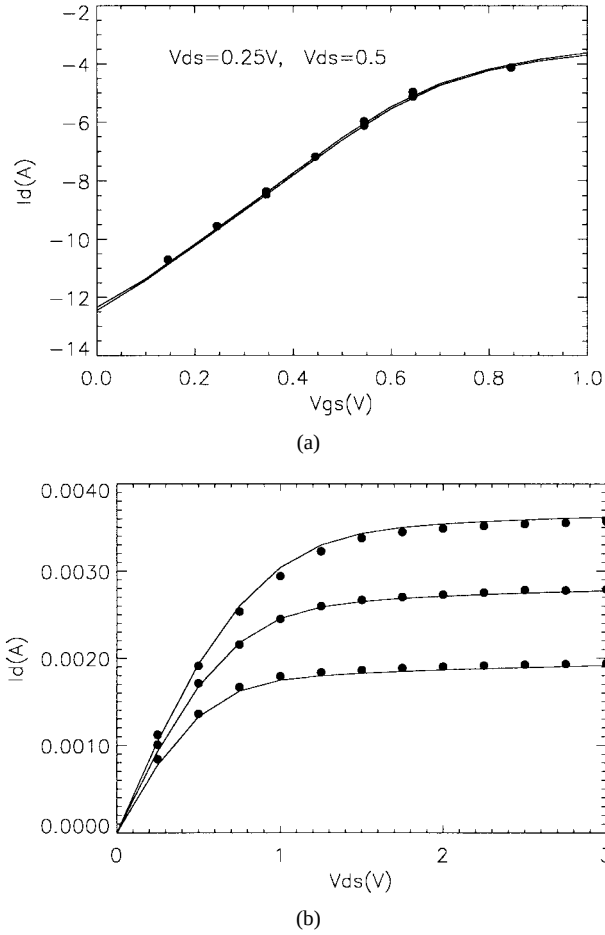


Fig. 8. Calculated and experimental I-V characteristics for the device for a wide range of applied biases. The dots are calculated by our simulator, whereas the solid lines are experimental values. (a) Agreement with experiment in the subthreshold region for a wide range of gate-source voltages with $V_{DS} = 0.25$ V and 0.5 V. (b) Agreement with experiment in the triode and saturation regions for a wide range of drain-source voltages with $V_{GS} = 2.0$ V, 2.5 V, and 3.0 V.

APPENDIX

In this Appendix, we express the collision integral in terms of the SH expansion technique for each of the important MOSFET scattering mechanisms. To do so, we place each of the scattering mechanisms into three major categories: 1) Inelastic Isotropic; 2) Elastic Isotropic; and 3) Elastic Anisotropic scattering.

A. Inelastic Isotropic Scattering

For the special case of inelastic isotropic scattering (iso), the transition probability S_{iso} is independent of angle. Therefore all coefficients in Legendre expansion (7), as well as all $1/\tau_{l,iso}^{in\mp}$, are zero except for $S_{0,iso}^{\mp}(\epsilon, \epsilon')$ and $1/\tau_{0,iso}^{in\mp}$. Equation (9) therefore reduces to the following general form for inelastic isotropic scattering:

$$\left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{iso} = \frac{f_0^0(\vec{r}, \epsilon \mp \Delta\epsilon)}{\tau_{0,iso}^{in\mp}(\epsilon)} Y_0^0(\theta, \psi) - \sum_{l,m} \frac{f_l^m(\vec{r}, \epsilon)}{\tau_{0,iso}^{out\mp}(\epsilon)} Y_l^m(\theta, \psi). \quad (16)$$

In this paper, we take optical and intervalley phonons, impact ionization, and electron-hole pair recombination to be isotropic-inelastic scattering, which we represent with the subscripts: op, ii, and rec, respectively.

1) *Optical and Intervalley Phonon Scattering*: We use the standard deformation potential approach for the matrix element [28] to find $\tau_{0,op}^{in\mp}$ and $\tau_{0,op}^{out\mp}$. Substituting this matrix element into (10), integrating, and inputting into (16) leads to expressions for the optical and intervalley phonon collision integrals which are identical to those found in [20]. (We note that in this paper, where applicable we employ the transport model put forth by [12], and do not distinguish between optical and intervalley phonons.)

2) *Impact Ionization*: The expressions for $1/\tau_{0,ii}^{out+}$ and $1/\tau_{0,ii}^{in\mp}$ have already been evaluated numerically, using a random- k approach, where agreement with experiment and Monte Carlo simulations was achieved [23]. Here, our generalized formulation allows us to directly input the resulting numerical values for $1/\tau_{0,ii}^{out+}$ and $1/\tau_{0,ii}^{in\mp}$. Reference [23] gives these values and details of how these values are obtained. To obtain the energy of the generated electron, we follow the work of [29], and divide the kinetic energy equally between the two secondary particles. While the scattering-in term has relatively little effect on the shape of the distribution function, it is included to maintain consistency.

3) *Electron-Hole Pair Recombination-Generation*: To adapt the Shockley-Hall-Read (SHR) recombination approach to the BTE, we begin along the lines of [30], and first revise the collision integral to account for the potentially degenerate distribution of electrons in traps

$$\begin{aligned} \left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{rec} &= \frac{V}{(2\pi)^3} \sum_{l,m} \int [f_{Tl}^m(\vec{r}, \epsilon') Y_l^m(\theta', \psi') S_{rec}(\vec{k}', \vec{k}) \\ &\quad - f_l^m(\vec{r}, \epsilon) Y_l^m(\theta, \psi) S_{rec}(\vec{k}, \vec{k}') \\ &\quad \cdot (1 - f_{Tl}^m(\vec{r}, \epsilon'))] d^3\vec{k}'. \end{aligned} \quad (17)$$

To formally determine the electron trap occupancy f_T , one would have to solve another BTE which describes carrier transport in traps. This, however, is very complicated and will perhaps be the subject of a future paper. For the present purposes, we use the result that most recombination occurs in traps close to the middle of the bandgap [31]. f_T thus becomes a function of a single energy, which is E_t , and can be factored out of the collision integral. Furthermore, within the context of the SHR formulation, f_T is isotropic. The collision integral therefore reduces to the following form:

$$\left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{rec} = -\frac{1 - f_T}{\tau_{0,rec}^{out+}} \sum_{l,m} f_l^m(\vec{r}, \epsilon) Y_l^m(\theta, \phi) + \frac{(f_T)_0^0 Y_0^0}{\tau_{0,rec}^{in-}(\epsilon)}. \quad (18)$$

For SHR recombination f_T is given by [32]

$$f_T = \frac{\tau_p n + p_1 \tau_{0,\text{rec}}^{\text{out},+}}{\tau_p(n + n_1) + \tau_{0,\text{rec}}^{\text{out},+}(p + p_1)} \quad (19)$$

where

$$n_1 = 2 \left(\frac{2\pi m_n^* K_B T_0}{h^2} \right)^{3/2} \exp \left(\frac{E_t}{K_B T_0} \right) \quad (20)$$

$$p_1 = 2 \left(\frac{2\pi m_p^* K_B T_0}{h^2} \right)^{3/2} \exp \left(-\frac{E_t + E_g}{K_B T_0} \right) \quad (21)$$

and $\tau_{0,\text{rec}}^{\text{out},+}$ and τ_p are interpreted to be the recombination lifetimes for electrons and holes, respectively [32].

We now proceed to determine a relationship between $\tau_{0,\text{rec}}^{\text{in}-}$ and $\tau_{0,\text{rec}}^{\text{out},+}$ so that detailed balance is satisfied. To achieve this, we start with the equilibrium condition

$$\left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{\text{rec}} = 0. \quad (22)$$

The equilibrium distribution f_{T0} for electrons in traps is given by

$$f_{T0} = \frac{1}{1 + \exp \left(\frac{E_t - E_f}{K_B T_0} \right)}. \quad (23)$$

From the equilibrium condition and (18) we have

$$\frac{1 - f_{T0}}{\tau_{0,\text{rec}}^{\text{out},+}(\varepsilon)} \sum_{l,m} f_l^m(\vec{r}, \varepsilon) Y_l^m(\theta, \phi) = \frac{(f_{T0})_0^0 Y_0^0}{\tau_{0,\text{rec}}^{\text{in}-}(\varepsilon)}. \quad (24)$$

By symmetry, this must reduce to the following:

$$\frac{1 - f_{T0}}{\tau_{0,\text{rec}}^{\text{out},+}(\varepsilon)} f_0(\vec{r}, \varepsilon) = \frac{f_{T0}}{\tau_{0,\text{rec}}^{\text{in}-}(\varepsilon)} \quad (25)$$

where

$$f_0(\vec{r}, \varepsilon) = \exp \left(-\frac{\varepsilon - E_f}{K_B T_0} \right) \quad (26)$$

is the Maxwell-Boltzmann distribution function of electron at equilibrium.

Now, by substituting f_{T0} and $f_0(\vec{r}, \varepsilon)$ into (25), we get the following relationship between: $\tau_{0,\text{rec}}^{\text{in}-}$ and $\tau_{0,\text{rec}}^{\text{out},+}$

$$\tau_{0,\text{rec}}^{\text{in}-}(\varepsilon) = \tau_{0,\text{rec}}^{\text{out},+} \exp \left(\frac{E_t - \varepsilon}{K_B T_0} \right). \quad (27)$$

Substituting the above expression for $\tau_{0,\text{rec}}^{\text{in}-}$ into (18), we obtain the following collision term for SHR recombination:

$$\begin{aligned} \left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{\text{rec}} &= -\frac{1 - f_T}{\tau_{0,\text{rec}}^{\text{out},\text{SHR}}(\varepsilon)} \sum_{l,m} f_l^m(\vec{r}, \varepsilon) Y_l^m(\theta, \phi) \\ &\quad + \frac{f_T}{\tau_{0,\text{rec}}^{\text{out},\text{SHR}}(\varepsilon)} \exp \left(\frac{\varepsilon - E_t}{K_B T_0} \right) \end{aligned} \quad (28)$$

where values for $\tau_{0,\text{rec}}^{\text{out},+}$ and τ_p are taken from the literature [32].

B. Elastic Anisotropic Scattering

For the special case of *elastic anisotropic* (ea) scattering, the change in energy $\Delta\varepsilon = 0$. Therefore, the coefficients f_l^m are the same for scattering-in and scattering-out, and can be factored out of the integral in (9). Furthermore, from (10) it can be seen that for the elastic case $\tau_{0,\text{ea}}^{\text{out}}(\varepsilon) = \tau_{0,\text{ea}}^{\text{in}}(\varepsilon)$. The zero order terms therefore cancel, and (9) reduces to the following generalized SH expression for elastic anisotropic scattering:

$$\left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{\text{ea}} = - \sum_{l=1,m}^{\infty} \frac{f_l^m(\vec{r}, \varepsilon)}{\tau_{l,\text{ea}}(\varepsilon)} Y_l^m(\theta, \phi) \quad (29)$$

where

$$\begin{aligned} \frac{1}{\tau_{l,\text{ea}}(\varepsilon)} &= \frac{1}{\tau_{0,\text{ea}}^{\text{out}}(\varepsilon)} - \frac{1}{2l+1} \frac{V}{(2\pi)^3} \\ &\quad \cdot \int S_{l,\text{ea}}(\varepsilon, \varepsilon') h(\varepsilon') d\varepsilon'. \end{aligned} \quad (30)$$

We note that in the above expression the superscripts have been eliminated because $1/\tau_{l,\text{ea}}$ reflects both scattering-in and scattering-out.

Ionized impurity scattering (imp), given by the Brooks-Herring (BH) model, falls under the elastic-anisotropic category. To obtain $\tau_{l,\text{imp}}$ one would project each of the Legendre polynomial basis functions onto the BH matrix element. This can be done by numerical integration. For our specific application, we take an analytical approach and formulate the ionized impurity collision term to first order in l . Analytical expressions for the $l = 0, 1$ terms are given in [19]. Using the expressions from [19] in (30), yields the following expression for the collision integral arising from ionized impurity scattering:

$$\left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{\text{imp}} = - \sum_{m=-1}^1 \frac{f_1^m(\vec{r}, \varepsilon)}{\tau_{1,\text{imp}}^{\text{out}}(\varepsilon)} Y_1^m(\theta, \psi) \quad (31)$$

where

$$\begin{aligned} \frac{1}{\tau_{1,\text{imp}}^{\text{out}}(\varepsilon)} &= \frac{\sqrt{2} Z^2 q^4 N_i}{64\pi^2 \epsilon_s^2 m^{*1/2}} \frac{\gamma'(\varepsilon)}{\gamma^{3/2}(\varepsilon)} \\ &\quad \cdot \left[\ln \left(\frac{4\gamma(\varepsilon) + a}{a} \right) - \frac{4\gamma(\varepsilon)}{4\gamma(\varepsilon) + a} \right] \end{aligned} \quad (32)$$

and

$$a = \frac{\hbar^2 \beta^2}{2m^*}; \quad \beta = \sqrt{\frac{4\pi q^2 n_{\text{scr}}}{\epsilon_s K_B T}} \quad (33)$$

where β is the inverse screening length, n_{scr} is the screening electron charge density, N_i the impurity concentration, K_B is the Boltzmann constant, T is the lattice temperature, ϵ_s is the permittivity of silicon, and Zq the ionized impurity charge.

C. Elastic Isotropic Scattering

For the elastic-isotropic (ei) case, all scattering-in terms with $l > 0$ are immediately eliminated due to orthogonality. Furthermore, as with elastic-anisotropic scattering, the zero order terms cancel. This results in the following generalized

SH expression for elastic isotropic scattering:

$$\left[\frac{\partial f(\vec{r}, \vec{k})}{\partial t} \right]_{\text{ei}} = -\frac{1}{\tau_{0,\text{ei}}^{\text{out}}(\varepsilon)} \sum_{l=1,m}^{\infty} f_l^m(\vec{r}, \varepsilon) Y_l^m(\theta, \psi). \quad (34)$$

We discuss below that bulk acoustic phonons, surface acoustic phonons, as well as surface roughness scattering fall into the category of elastic isotropic scattering.

Acoustic phonons are just a few meV, so we take them to be elastic. The elastic approximation leads to a matrix element for acoustic phonons which is elastic-isotropic [28]. Using the above expression for elastic-isotropic scattering, and substituting the matrix element into (10), yields the following expression for $1/\tau_{0,\text{ac}}^{\text{out}}$

$$\frac{1}{\tau_{0,\text{ac}}^{\text{out}}(\varepsilon)} = \frac{2^{1/2} m^{*3/2} D_{\text{ac}}^2 K_B T}{\pi \rho \hbar^4 u^2} \gamma^{1/2}(\varepsilon) \gamma'(\varepsilon) \quad (35)$$

where D_{ac} is the acoustic phonon deformation potential; u is the velocity of sound, and ρ is the density of silicon.

In MOSFET inversion layers carrier transport is different from motion in the bulk due to surface scattering and quantum confinement. To account for inversion-layer transport, we follow the work of Lombardi and co-workers, who obtained good results for drift-diffusion modeling by explicitly relating the scattering rate in the inversion layer to the field perpendicular to the Si-SiO₂ interface [33]. In this paper, we adapt the aforementioned method [33], to account for the effects of transport along the interface, by incorporating 2-D surface acoustic phonon scattering, as well as surface roughness scattering, in the SH technique for solving the BTE.

Surface acoustic (sa) phonons have been described by 2-D deformation potential theory [26]. As with the bulk case, we make the elastic approximation which gives rise to a matrix element which is isotropic. Integrating this matrix element over the 2-D density of states for one subband yields the following value for the surface acoustic phonon scattering rate: $1/\tau_{0,\text{sa}}^{\text{out}}$

$$\frac{1}{\tau_{0,\text{sa}}^{\text{out}}} = \frac{m^* D_{\text{sa}}^2 K_B T}{\hbar^3 u^2 \rho_A} \quad (36)$$

where ρ_A is the areal density, and D_{sa} is the deformation potential. To account for the normal electric field and additional quantum effects in surface acoustic scattering, we further adopt the arguments of Lombard and Schwarz [33], [34]. In their works, the authors express the areal density to be equal to the bulk density times the average width of the inversion layer, $\rho_A = \rho Z_{\text{avg}}$ [33], [34]. Using classical arguments, as well as the Heisenberg uncertainty principle, the authors argued that Z_{avg} is inversely proportional to the normal electric field. This gives rise to the following expression for $1/\tau_{0,\text{sa}}^{\text{out}}$

$$\frac{1}{\tau_{0,\text{sa}}^{\text{out}}} = \left\{ \frac{\hbar^3 u^2 \rho}{q m D_{\text{sa}}^2 K T} \left[\frac{3 K T}{2 E_{\perp}} + \frac{\delta \hbar^{2/3} q^{2/3}}{m^{*1/3} E_{\perp}^{1/3}} \right] \right\}^{-1} \quad (37)$$

where the second term in (37) reflects quantum effects of a triangular potential well, and δ is an empirically determined parameter which gives an indication of the strength of the quantum term.

To account for *surface roughness (sr) scattering*, we again adopt the arguments of [33] by first approximating the inversion layer as an ideal triangular potential well. Under this approximation, the surface roughness scattering rate $1/\tau_{0,\text{sr}}^{\text{out}}$ is usually taken to be proportional to the square of the surface carrier concentration n_s and the depletion region areal charge N_{dr}

$$\frac{1}{\tau_{0,\text{sr}}^{\text{out}}} = \frac{m^* \Delta^2 L^2 q^4}{2 \hbar^3 \epsilon^2} (n_s + N_{\text{dr}})^2 f_{\text{sr}} \quad (38)$$

where Δ and L are the rms height and correlation length of the roughness respectively; and f_{sr} is a slowly varying function of energy [27].

Using Gauss' law, we now relate the surface charge $n_s + N_{\text{dr}}$ to the normal electric field $n_s + N_{\text{dr}} = E_{\perp} \epsilon_s / q$. The rate $1/\tau_{0,\text{sr}}^{\text{out}}$ can now be expressed in the following form:

$$\frac{1}{\tau_{0,\text{sr}}^{\text{out}}} = \Gamma E_{\perp}^2 \quad (39)$$

where all the above physical parameters have been lumped into Γ .

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REFERENCES

- [1] M. A. Alam, M. A. Stettler, and M. S. Lundstrom, "Formulation of the Boltzmann equation in terms of scattering matrices," *Solid-State Electron.*, vol. 36, p. 263, 1993.
- [2] F. J. Mustieles and F. Delaurens, "Numerical simulation of nonhomogeneous submicron semiconductor devices by a deterministic particle method," *Solid-State Electron.*, vol. 36, p. 857, 1993.
- [3] C. C. C. Leung and P. A. Childs, "Spatially transient hot-electron distributions in silicon determined from the chambers path integral solution of the Boltzmann transport equation," *Solid-State Electron.*, vol. 36, p. 1001, 1993.
- [4] X. L. Wang, S. Jallepalli, C. F. Yeap, and C. M. Maziar, "Two-dimensional characteristic method for Boltzmann transport equation and its application to deep submicron MOSFET's," in *SRC TECHCON'93*, 1993, pp. 352-354.
- [5] M.-C. Cheng and R. Chennupati, "Evolution of nonequilibrium electron distribution functions at hydrodynamic scales in multivalley semiconductors," *J. Phys.*, vol. 28, pp. 160-173, 1995.
- [6] A. Forghieri, R. Guerrieri, P. Ciampolini, A. Gnudi, M. Rudan, and G. Baccarani, "A new discretization strategy of the semiconductor equations comprising momentum and energy balance," *IEEE Trans. Computer-Aided Design*, vol. 7, pp. 231-242, 1988.
- [7] B. Meinerzhagen and W. L. Engl, "The influence of the thermal equilibrium approximation on the accuracy of classical two-dimensional numerical modeling of silicon submicrometer MOS transistors," *IEEE Trans. Electron Devices*, vol. 35, pp. 689-697, 1988.
- [8] K.-W. Chai, P. Mawby, and A. McCowen, "Hydrodynamic simulation of electron heating in conventional and lightly-doped-drain MOSFET's with application to substrate current calculation," *Int. J. Numer. Model.: Electronic Networks, Devices, and Fields*, vol. 5, pp. 53-66, 1992.
- [9] D. Chen, E. C. Kan, U. Raviooli, C.-W. Shu, and R. W. Dutton, "An improved energy transport model including nonparabolicity and non-Maxwellian distribution effects," *IEEE Electron Device Lett.*, vol. 13, pp. 26-28, 1992.
- [10] Q. Lin, N. Goldsman, and G.-C. Tai, "Highly stable and routinely convergent two-dimensional hydrodynamic device simulation," *Solid-State Electron.*, vol. 37, pp. 359-371, 1994.
- [11] W.-S. Choi, J.-G. Ahn, Y.-J. Park, H.-S. Min, and C.-G. Hwang, "A time dependent hydrodynamic device simulator snu-2d with new discretization scheme and algorithm," *IEEE Trans. Computer-Aided Design*, vol. 13, pp. 899-908, July 1994.

- [12] C. Fiegna and E. Sangiorgi, "Modeling of high-energy electrons in MOS devices at the microscopic level," *IEEE Trans. Electron Devices*, vol. 40, pp. 619–627, March 1993.
- [13] R. Brunetti and C. Jacoboni, "A many-band silicon model for hot-electron transport at high energies," *Solid-State Electron.*, vol. 32, pp. 1663–1667, 1989.
- [14] N. Goldsman, L. Henrickson, and J. Frey, "A physics-based analytical/numerical solution to the Boltzmann transport equation for use in device simulation," *Solid-State Electron.*, vol. 34, pp. 389–396, 1991.
- [15] Y.-J. Wu and N. Goldsman, "An efficient solution of the multi-band Boltzmann transport equation in silicon," *COMPEL*, vol. 12, pp. 475–485, 1993.
- [16] H. Lin, N. Goldsman, and I. D. Mayergoyz, "An efficient deterministic solution of the space-dependent Boltzmann transport equation for silicon," *Solid-State Electron.*, vol. 35, pp. 33–42, 1992.
- [17] A. Gnudi, D. Ventura, and G. Baccarani, "Modeling impact ionization in a BJT by means of spherical harmonics expansion of the Boltzmann transport equation," *IEEE Trans. Computer-Aided Design*, vol. 12, p. 1706, 1993.
- [18] H. Lin, N. Goldsman, and I. D. Mayergoyz, "Device modeling by deterministic self-consistent solution of Poisson and Boltzmann transport equations," *Solid-State Electron.*, vol. 35, pp. 769–778, 1992.
- [19] A. Gnudi, D. Ventura, G. Baccarani, and F. Odeh, "Two-dimensional MOSFET simulation by means of a multidimensional spherical harmonics expansion of the Boltzmann transport equation," *Solid-State Electron.*, vol. 36, p. 575, 1993.
- [20] K. A. Hennacy, Y.-J. Wu, N. Goldsman, and I. D. Mayergoyz, "Deterministic MOSFET simulation using a generalized spherical harmonic expansion of the Boltzmann equation," *Solid-State Electron.*, vol. 38, pp. 1498–1495, 1995.
- [21] K. A. Hennacy and N. Goldsman, "A generalized legendre polynomial/sparse matrix approach for determining the distribution function in nonpolar semiconductors," *Solid-State Electron.*, vol. 36, pp. 869–877, 1993.
- [22] G. B. Arfken, *Mathematical Methods for Physicists*. New York: Academic, 1966.
- [23] Y.-J. Wu and N. Goldsman, "Deterministic modeling of impact ionization with a random- k approximation and the solution of the multiband Boltzmann transport equation," *J. Appl. Phys.*, vol. 78, 1995.
- [24] C. E. Korman and I. D. Mayergoyz, "A globally convergent algorithm for the solution of the steady-state semiconductor device," *J. Appl. Phys.*, vol. 68, p. 1324, 1990.
- [25] N. Khalil, J. Faricelli, D. Bell, and S. Selberherr, "A novel method for extracting the two-dimensional doping profile of a sub-half micron MOSFET," in *1994 Symp. VLSI Technol. Dig. Tech. Papers*, 1994, pp. 131–132.
- [26] C. Sah, T. Ning, and L. Tschopp, "The scattering of electrons by surface charges and by lattice vibrations at the silicon-silicon dioxide interface," *Surf. Sci.*, vol. 32, pp. 561–575, 1972.
- [27] S. Goodnick, R. Gann, D. Ferry, and C. Wilmsen, "Surface roughness induced scattering and band tailing," *Surf. Sci.*, vol. 113, pp. 233–238, 1982.
- [28] M. Lundstrom, *Fundamentals of Carrier Transport, Modular Series on Solid-State Devices*. New York: Addison-Wesley, 1990.
- [29] P. Landsberg, "Impact ionization and auger recombination in band," *Solid-State Commun.*, vol. 10, pp. 479–482, 1972.
- [30] K. Hess, *Advanced Theory of Semiconductor Devices*. Englewood Cliffs, NJ: Prentice-Hall, 1988.
- [31] R. Muller and T. Kamins, *Device Electronics for Integrated Circuits*. New York: Wiley, 1986.
- [32] S. Selberherr, *Analysis and Simulation of Semiconductor Devices*. Wein, New York: Springer-Verlag, 1984.
- [33] C. Lombardi, S. Manzini, A. Saporito, and M. Vanzi, "A physically based mobility model for numerical simulation of nonplanar devices," *IEEE Trans. Computer-Aided Design*, vol. 7, pp. 1164–1171, 1988.
- [34] S. Schwarz and S. Russek, "Semi-empirical equations for electron velocity in silicon: Part II—MOS inversion layer," *IEEE Trans. Electron Devices*, vol. ED-30, pp. 1634–1639, 1983.



Wenchao Liang received the B.S. degree in physics from Tsinghua University, Beijing, China, in 1989, and the M.S. and Ph.D. degrees in electrical engineering and from the University of Maryland, College Park, in 1994 and 1996, respectively.

From 1991 to 1992, he was a Teaching Assistant in the Department of Physics, University of Maryland. He held an RA position working on III–V compound semiconductor quantum devices from 1992 to 1993, and a TA position from 1993 to 1994 in the Department of Electrical Engineering at the same university. Currently, he is with Digital Equipment Corporation, Hudson, MA.

Neil Goldsman received the Ph.D. degree in electrical engineering from Cornell University, Ithaca, NY, in 1989.

Currently, he is an Associate Professor at the University of Maryland, College Park. His main research interests are in the areas of device and material modeling. He has published approximately 70 journal and conference papers.

Dr. Goldsman is the recipient of the National Science Foundation's Research Initiation Award, the University of Maryland IEEE Professor of the Year Award, and the George Corcoran Award for Contributions to Education.

Isaak Mayergoyz (SM'83–F'88) received the Ph.D. degree in electrical engineering from the Cybernetics Institute of the Ukrainian Academy of Sciences, Kiev, Ukraine.

Currently, he is a Professor in the Department of Electrical Engineering, University of Maryland, College Park, and has been a member of the university's faculty since 1981. His research interests are in the areas of magnetics, computational electromagnetics, applied superconductivity, numerical modeling of semiconductor devices, power engineering, and applied mathematics. He has published three scientific books and more than 170 scientific papers.

Dr. Mayergoyz is a Research Fellow of GE Corporate Research and Development, a Distinguished Lecturer of the Magnetic Society, and has served for three years on the Edison Medal Committee (IEEE Award Board). He is a recipient of the Outstanding Teaching Award of the College of Engineering and the Distinguished Scholar-Teacher Award at the University of Maryland.

Phil J. Oldiges received the B.S. degree in physics from Thomas More College, Fort Mitchell, KY, in 1981, and the M.S. and Ph.D. degrees in electrical engineering from Cornell University, Ithaca, NY, in 1984 and 1988, respectively.

From 1984 to 1986, he was a Visiting Research Scientist at Toshiba Corporation, Kawasaki, Japan, where he investigated the dynamics of alpha-particle induced soft errors in dynamic memories. From 1988 to 1993, he worked at Sony Corporation, Atsugi, Japan, in the area of TCAD applications to advanced silicon device technologies. Currently, he is a member of the TCAD group at Digital Semiconductor, Hudson, MA, working in the area of TCAD semiconductor process simulation and the modeling of alpha-particle effects in semiconductor devices and circuits.